

Kailo Coaching Hub Audit

Product review & data-platform architecture

21 August 2026

Defining Product Goals

- **Team-native, coach-facing training platform.** The team is the unit; the coach is the primary user.
- **Stronger analytical tools and coach customization** than TrainingPeaks or FinalSurge — the analytics and the ability to tailor the surface are the differentiators, not feature parity.
- **A coach-determined path for athletes to interact** that *reduces* the need for athlete buy-in. The less work an athlete has to do, the better the product: participation should be near-passive by default, with the coach deciding how much the athlete is asked to do.
- **Highly customizable.** Coaches change the information they see, where it sits, and what it is called — the layout is theirs, not a template they work around.
- **Expandable and collapsible.** The system is iterative: a coach can run it as a roster and a workout planner and nothing else, or turn on every analytic layer, and move between those states at will. Some coaches will want all of it and some almost none; both are first-class users.

Next Steps (ordered)

1. **Restructure the hub into the five-tab layout** of §1.3 — Today, Upcoming, Team, Recruiting (v2), Settings — with the athlete page reached through the Team tab (roster, results, or the PR / progression boards) rather than as its own tab.
 2. **Athlete objects with histories and attributes.** Each athlete carries a history (race and training paces over time) *and* a set of learned attributes — e.g. “performs better on hilly courses,” “degrades more in rain / heat.” These attributes are estimated from data, they change as more data arrives, and the set will grow. The data platform (§2) exists to populate and keep refreshing them.
 3. **Text-to-structured-workout parsing.** The planner grid must turn “2 mile wu, 5 tempo, 3 cd” into a structured workout that can be analyzed post hoc; the current parser misreads such inputs (see the Planner and Workout planner audits).
 4. **Weather upgrade.** Move the adjustment metric from dew point to wet-bulb globe temperature, and show conditions for four windows — this morning / afternoon, next morning / afternoon — to cover doubles.
 5. **Data & enrichment platform build-out** (§2): divest from Athletic.net; per-race time, weather, and hill enrichment feeding the learned attributes.
 6. **Recruiting** — a *v2 product*. Reserved in the tab structure but not built in v1.
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1 Product Audit

1.1 Two axes

The desktop hub is organized on two axes.

- **Vertical axis — thematic.** A left pane (top on narrow screens) of **five tier-0 objects**, each containing pages (Figure 1). The tier-0 objects are *Before Practice*, *Share with the Team*, *After Practice*, *The Roster*, and *Housekeeping*. Their labels are not clickable; the pages beneath them are. This axis is the spine of the audit below — every page gets an analysis, and every tier-0 object gets a general assessment.

BEFORE PRACTICE	SHARE WITH THE TEAM	AFTER PRACTICE	THE ROSTER	HOUSEKEEPING
Today	Athlete day	Session review	Week wall	Operations
Tasks + Links	Team board	Inbox	Volume	Data health
Planner	1-on-1s		Athlete	How this works P7
Workout planner			Parameters	Setup
Pace cards			Results	
			Predictions	

Figure 1: Vertical (thematic) axis as of August 21, 2026

- **Horizontal axis — temporal.** The coach's day, as a left-to-right flow. Each step is simply a **hyperlink** to one of the vertical-axis pages (Figure 2).

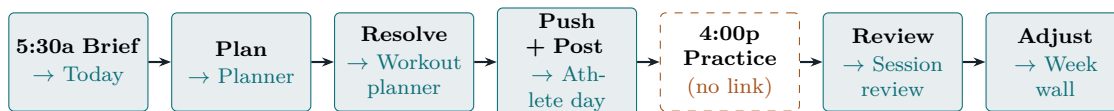


Figure 2: The horizontal (temporal) axis: the coach's day. Each step links to a vertical-axis page. **4:00p Practice currently links nowhere.**

The day cuts across the thematic tiers. The seven day-steps do not live inside one tier-0 object — they thread through four of them:

Day step	Links to	Tier-0 home
5:30a Brief	Today	Before Practice
Plan	Planner	Before Practice
Resolve	Workout planner	Before Practice
Push + Post	Athlete day	Share with the Team
4:00p Practice	— (nothing)	—
Review	Session review	After Practice
Adjust	Week wall	The Roster

[Future consideration should be made to decide whether information should be organized in a temporal or thematic axes.]

1.2 Vertical axis — page-by-page audit

Before Practice

Today [← day: 5:30a Brief]

This is the landing page ([kailo.fit/hub](#)) — a long stack of text widgets. Top widgets: “[N] uploaded yesterday,” “Available today,” “Data flags,” “Gone quiet,” “Need a look,” and “Dew point at practice.” Below them: “Where your attention goes” (activities flagged abnormal — average heart rate high on an easy run, distance longer than expected), a “Today’s plan” widget with a print option, and a practice conditions widget (“Conditions this afternoon”).

- **Keep high-quality, accessible print.** The plan’s print option is good and should stay easy to reach throughout.
- **Conditions is buried and single-window.** It sits too far down, does not currently populate, and only shows the afternoon — athletes running *doubles* lose the morning window entirely.
- **No expand/collapse** on any widget, producing a very long page.

Assessment. Too much information, almost all of it text, with no visual hierarchy to separate what matters. Because this is the landing page, deep customization is essential.

Tasks + Links

This is a simple and considerably shorter page than the Today screen. It consists of two widgets: a basic task organizer and a place to store hyperlinks. The quick links widget prompts, “Add the ones you reach for weekly - entries, results, the conference site.”

Assessment. A dedicated task page is likely useful, with a connected widget that can be added to a landing page. I don’t see much long-term need for the “Quick links” section. Anything that would go there should be put in its proper place.

Planner [← day: Plan]

The planner page is a core aspect of the product. It has planning capabilities for the entire season (day, week, and entire training block). Weeks can be toggled one by one. There is a pin every Wednesday in this example. I am interested to know how the week grid works. Namely, does inputting text create a formatted workout? Ideally, one could write “Easy 45 minutes” or “2 mile wu, 5 tempo, 3 cd” in the grid and then recover an analysis of the workout based on how it was completed vs. predicted (easy pace was faster or higher-HR than normal; tempo pace is improving).

Assessment. This will be a core page going forward that should anchor a future “Upcoming” section. Technical considerations must be made to determine the degree to which text inputs can create structured workouts that can be analyzed post hoc.

Workout planner [← day: Resolve]

This seems to be a more zoomed-in version of the planner day above. It allows for creating workout pods, prescribing paces for a workout, and making weather/terrain adjustments. Again, there is some instructional language here that is just a bit over the top, such as “The question: one date, six lanes. I plug the workout in, the groups and paces come out, weather and terrain already applied.” I’m not sure what the “What moved since last week” section corresponds to, unless it’s simply product updates. There is a “Rules baked in” section, which is basically settings.

Assessment. Dew point is currently the listed weather metric used to determine weather adjustments. Based on the literature, we'd probably want this to move to wet-bulb globe temperature. The "Rules baked in" section should be removed and replaced with a link to the workout planner settings page, for organizational purposes. As above, we need to put in technical work to make the text-to-planner conversion work better. For instance, I tried to write "5 up, 2 tempo, 3 cd," and the current iteration put "2 tempo, 3 cd" as the recovery.

■ Pace cards

Basically the VDOT training paces. It is unclear what "Provisional" means here. It asks users whether they want "page" or "phone" views. The prescribed paces are: "This week," "Easy days," "Long run," "Weekly probe."

Assessment. This is again a core feature that should probably be linked in the training section but live under the Athlete pages. I don't understand the logic for why certain paces are excluded — namely recovery and tempo/LT.

Tier-0 assessment. In general this tier is too much of a catchall for the core elements of the product. This isn't necessarily a problem on its own. However, given how loaded this part is with features compared to the next two Tier-0 categories (Share with the Team and After Practice), it suggests there are better ways to organize these data and features to disperse them in a more coherent and balanced way. I would recommend creating a "Today" landing page and "Upcoming" planning sections that split up this data.

Share with the Team

■ Athlete day [[← day: Push + Post](#)]

This page allows you to see what a given athlete sees on their Kailo app for each day's practice.

Assessment. This page can likely be a submodule of existing planners as a "Preview" option.

■ Team board

This page is a grab bag of different planning and team-related tools. "This week" is the first tab and is similar to the Workout planner. PR Board, currently unpopulated, seems to be a way to assemble PRs from a given meet (or overall PRs — it is unclear which). The team document, also currently unpopulated, seems to be a place to record team rules/creeds/values. Race plans allow the user to add different goals/plans for specific athletes for specific races. You can also input other teams' results.

Assessment. This page has too many disparate elements that limit its usefulness. While there is a conceptual core, there are better places for race info, weekly planning, team rules, and team PRs. A specific team homepage would be useful, which could contain important division/conference information along with the team document. Race plans can definitely be improved by adding both predictive modeling for a given athlete (predicted pace range) and MCMC simulations to see how that athlete stacks up against expected competition.

■ 1-on-1s

This is a central repository to prepare and document 1-on-1s. The structure of the meeting notes isn't great since there is no expand/collapse, so long notes would render the page extremely long.

Assessment. I think this is probably a service that only some coaches will use, given that many won't input notes on meetings, but it is a good feature to have. However, like other features in this section, I think it could be moved to a better location.

Tier-0 assessment. Generally, the features in this tier are all individually useful, but some read as redundant with other features, and all of them could be placed in more intuitive locations. I would recommend removing this as an organizing tier and splitting its data among future Today, Upcoming, and Team sections.

After Practice

Session review [[← day: Review](#)]

Currently this page is not populated at all, which makes it hard for me to fully assess. But it seems to be information about the most recent session.

Assessment. This page is, of course, extremely central to coaching. We can create a “Most recent session results” module for the “Today” screen with the information in this section. But we need a space to put historical session data (e.g., the session from five days ago).

Inbox

Some of these are repeated widgets from the current “Today” page (namely the “Needs a look” section, with some overlap in “Contact coverage”). The Reason ledger and Confounds sections are not fully populated for me, so I can't fully assess what's in them (though the latter seems to be some predictive or analytical system).

Assessment. Similar to the Share with the Team section, I think there is important information in this section that would be better served by being moved under a different header in a more optimal organizing strategy.

Tier-0 assessment. I echo what I said in the Share with the Team section: many of these features are useful but are either redundant or not optimally placed. I also think that coaches need a way to toggle certain features on or off, such as meeting tracking with students. If a coach doesn't manually input that information, the app may nag them that they're never meeting with students. As part of the customization layer, it would be useful to treat some of the functionality like a grand-strategy video game: add layers/features as you want, or simply use it as a more basic tool of roster visuals and workout planning.

The Roster

Week wall [[← day: Adjust](#)]

This seems to be another calendar overview of the training week. For me, it is populated by only two athletes.

Assessment. Important information, but redundant with other pages. Perhaps other pages on training programs could have various view options, so you could have a “week view” option like this.

Volume

A key page. Athlete name, 7-day and 28-day volume with wk/wk %, rest days, last upload, percent complete, directive, loop, and trend (in histogram form) all sit under the default “This

window” pane. The second pane is “Plan & Goals,” which has weekly volume goals and season peak-mileage goals (and down-week depth). There are unpopulated Actuals and Adherence panes.

Assessment. There needs to be a legend for certain terms or concepts such as Complete %, wk/wk %, directive, and loop. The trending inlay graph is pretty, but I’m not sure about its use. In general, lots of the information here is useful. But again, the Plan & Goals section would be better nestled in a planning section.

Athlete

This is basically a roster. It is a drop-down. Upon selecting an athlete, you get “The person file” with high-school info, dietary restrictions, and class schedules. There is also information about surveys and check-ins, notes, and the athlete’s stated goals. The page continues to Paperwork, Training room, Recruiting file, Class schedule, and Medical and emergency contact.

Assessment. In general, this page is pretty long. This is more like an athlete’s settings than a landing page. A more optimal landing page would be closer to a public roster: name, grade, gender, hometown, PRs, recent race results, and recent training (a week-over-week mileage graph, perhaps), along with some future-oriented things like goals. Dietary restrictions and class schedules could still be populated below, but some of the things currently on the page could simply be links to other sections instead of repeated widgets (such as 1-on-1 notes).

Parameters

This is a list of the parameters produced through modeling for each athlete. These include critical speed, Rx width, D-prime, anchors, speed reserve, and extrapolation.

Assessment. The listed critical speed (with the CIs attached) could be an optional widget added to athlete profiles. This page, generally, is probably best kept in the settings.

Results

As expected, a results page with tabs for “by meet,” “season progression,” “record board,” “courses,” “import,” “goals,” and “season review.”

Assessment. As said above, some of the recent results and PRs could probably be included as (optional) widgets on the athlete profile page. But a central results repository is still important to include in the “Team” section. The import option should be a hyperlink to a pop-up, not its own tab. Some of the remaining information could likely be either reorganized or broken into different pages.

Predictions

This is a page devoted to assessing how accurate past race predictions were. Past results are shown (“Locked before the gun”) in a widget, followed by a grading widget, a calibration widget, and a larger season replay widget.

Assessment. I’m not sure how central this page is. Perhaps because no predictions have been made yet — and I can’t see current predictions — this page seems better nestled in the settings.

Tier-0 assessment. This is the section where not being able to click the tier-0 module name (The Roster) is most frustrating. I think the optimal flow would be a “Team” landing page with team name, picture, and conference/division information. This would then have a button for the roster, another for results, another for PRs, and perhaps season progression.

Housekeeping

Operations

This is a page for logistical/operational components of coaching, such as money or equipment information. It also allows .ics export of meet/training calendar information. The first two tabs — season board and this meet — seem to be repeats of existing data.

Assessment. This is probably better nestled under the Team label instead of the Housekeeping label, since I think the latter is better served as settings.

■ Data health

This is more of a classic settings page on data importation. The first widget (Coverage) shows runs carrying a heart-rate average, session uploads carrying lap marks, and athletes with a band prescribed today. This is followed by a widget on whose data is not arriving. There is then a long widget of where each athlete's data comes from (Strava vs. Garmin). There is also an extremely long section of flags from the data ("Things worth knowing about the data"). Finally, there is a widget showing who is missing paperwork.

Assessment. I'm not sure whether the paperwork described is medical clearances. If so, that information is too different from the data stuff here to be included. I think this could be recast as a "Data importation" settings menu.

■ How this works

An unpopulated introduction/explainer for the app.

Assessment. Obvious need.

■ Setup

Lots of settings panels here, which can all be broken into their own settings tabs. These include "Recruit-visit mode," which masks athlete names "so a campus visit sees the machinery and not the people." There are also settings about program information, team information (name, location, coaches/access settings), and data flags. There are also bulk import and export widgets.

Assessment. These need to be built out as their own settings menu. This includes recruit-visit mode, team settings (including team/practice location), coach settings, bulk import/export, and the various other settings listed above.

Tier-0 assessment. There is a clear need to make this tier into a more standardized settings menu that centralizes many of the options scattered throughout the current system.

1.3 Proposed structure: five main tabs

Collapse the sprawl into five tabs. This is a first pass, assembled from the per-page recommendations above.

1. **Today** — the landing page. Keeps what the current Today page gets right (the plan, with the print option easy to reach) and adds what the audit calls for: a "most recent session" module (from Session review), the needs-a-look flags, and practice conditions for *four* windows — this morning, this afternoon, next morning, next afternoon (fixing the doubles gap). Every widget collapsible, and the widget set coach-customizable.
2. **Upcoming** — planning, anchored by the Planner (day / week / training block) with the Workout planner as its zoomed-in day view. Absorbs race plans (from Team board) and the Plan & Goals material currently inside Volume.

3. **Team** — a team landing page (name, picture, conference/division information, the team document), with routes into:
 - **Roster** — the public-roster-style list; selecting an athlete opens their page (layout below);
 - **Results** — the central results repository (by meet, season progression, record board, courses), where every athlete name links to that athlete’s page;
 - **PRs / season progression** — selecting an athlete goes to their page;
 - **Operations** — logistics: money, equipment, the .ics calendar export.
- The athlete page is deliberately *not* a top-level tab: the routes converge on it (Figure 7), so a coach reaches the same athlete object whether they come through the roster, the boards, or a result.
4. **Recruiting** — reserved as a tab, but a *v2 product*; not built in v1. (Recruit-visit mode — masking athlete names — ships in v1 under Settings.)
 5. **Settings** — the standardized menu the Housekeeping audit calls for: data importation (Data health), team / coach / program settings, the workout planner rules, Parameters and Predictions, bulk import/export, recruit-visit mode, and the feature toggles (turn modules such as 1-on-1 tracking on or off).

Table 1 maps the proposal onto the current hub; the expanded plans below give the per-tab build detail.

Tab	Routes / contents	Absorbs from current hub
Today	glance row; conditions (four windows); today’s plan + print; needs a look; most recent session; tasks	Today, Tasks + Links, Session review (as a module), Inbox (flags)
Upcoming	the planner at four zooms: block → week → day → athlete preview; race plans; plan & goals	Planner, Workout planner, Athlete day, Team board (race plans), Volume (goals), Week wall (as a view)
Team	landing page; Roster; Results; PRs / progression; Operations; the athlete pages	The Roster tier, Team board (team document, PR board), Operations, 1-on-1s (linked from athlete pages)
Recruiting	v2 placeholder	Recruiting file (from the person file)
Settings	data import; team / coach / program; workout rules; parameters; predictions; feature toggles; recruit-visit mode; bulk import/export; help	Data health, Setup, Parameters, Predictions, “Rules baked in,” How this works

Table 1: The proposed five-tab structure, and where the current hub’s pages land.

1.4 Expanded plans

First-pass build detail per tab. The athlete page gets the most, since it is the object the rest of the product hangs off.

Today

The landing page answers “what do I need to know before practice” in one screen. Figure 3 is the *default* layout — every slot on it is a pick from the widget catalog below, so two coaches’ Today

pages need not look alike.

- **Greeting line** — “Good morning, Coach. Today is Tuesday, September 8 · XC day 34.” Time-of-day aware. The salutation, the name, and what follows the date (season day count, days to the next race, nothing) are all fields the coach sets. The day count is *sport*-specific — “XC day 34,” “Indoor day 12,” “Outdoor day 40” — never a generic day number, since a program runs three seasons a year.
- **Glance row** — four small slots. The default four are uploaded yesterday, athletes available today, data flags / gone quiet, and needs a look; any slot can be swapped for another glance widget — days until the next race, paperwork outstanding, team mileage against plan (Table 2).
- **Conditions strip** — four windows (this morning, this afternoon, tomorrow morning, tomorrow afternoon) so doubles are covered. The *model* behind workout adjustments is wet-bulb globe temperature, but WBGT is a poor thing to show a coach. Each window displays temperature, a conditions icon (sun / cloud / rain), precipitation chance, and humidity; the WBGT-derived flag surfaces only as the window’s colour.
- **Today’s plan** — the print option kept one click away.
- **Most recent session** — the Session review material as a module, with historical sessions reachable from it.
- **Tasks** — the task organizer from Tasks + Links as a widget.

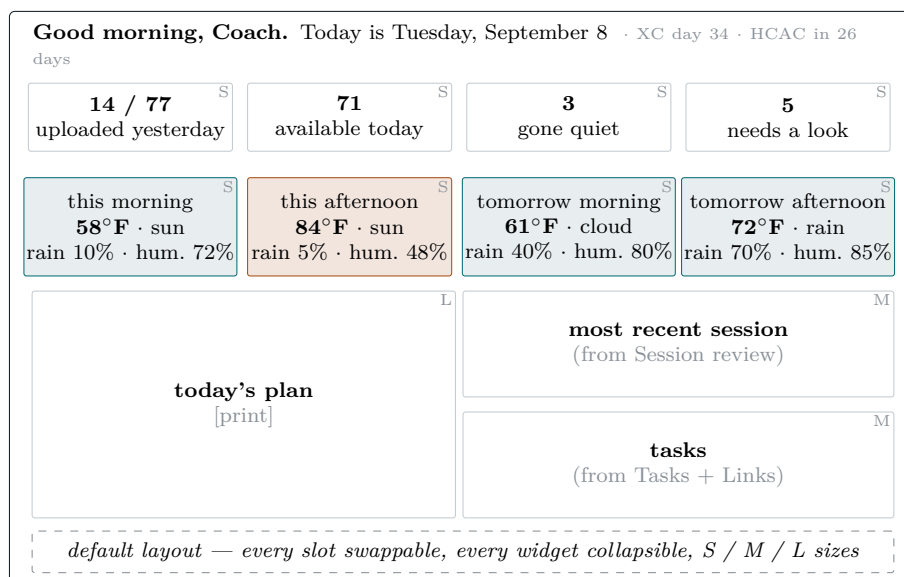


Figure 3: Today, default layout. The glance row is four swappable slots; the conditions strip shows what a coach reads (temperature, icon, rain, humidity) while WBGT drives only the window’s colour — the afternoon here is flagged. Corner letters are widget sizes.

Widgets come in three sizes. Following the Apple widget model: a **glance** (S) widget is one number or flag in a 1×1 tile; a **medium** (M) widget is 2×1 and holds a short list or a compact chart; a **large** (L) widget is 2×2 and holds a real chart or a rich panel. The page is a grid the coach arranges — add, remove, reorder, and resize where a widget supports two sizes. Some widgets

Glance widget (S)	Shows	Type
Uploaded yesterday	n of N athletes with a synced activity	count
Available today	athletes cleared to practice	count
Data flags / gone quiet	athletes with no upload in k days	count, links
Needs a look	activities flagged abnormal (HR high on easy, distance off plan)	count, links
Days until next race	countdown plus the meet name	number
Season day	“XC day 34 of 112” — counted within the current sport season (XC / indoor / outdoor), or weeks to the championship	number
Injured / training room	athletes on a modified plan	count, links
Paperwork outstanding	missing clearances	count, links
Team mileage vs. plan	this week’s volume as a percent of planned	percent
Check-ins missing	surveys or check-ins not returned	count
Conditions flag	today’s WBGT band as a colour	flag
Entry deadline	next meet’s entry cutoff	date
Readiness alerts	sleep / HRV outside the athlete’s band (when connected)	count
Birthdays this week	—	short list

Table 2: Glance widgets: candidates for the four small slots.

are text, some are numbers, some are charts; the catalog marks which, and the point of the chart widgets is that the coach never has to assemble a graph by hand.

Build order: the widget framework (grid, collapse, persistence) first, since everything else on the page is a widget; then the conditions strip; then the recent-session module; then the catalog grows one widget at a time, charts last.

Upcoming

One planning object at four zoom levels, rather than three separate pages (Figure 5): the season block (phases, goals, peak mileage, down-week depth — absorbing Volume’s Plan & Goals), the week grid (Week wall becomes a view option here, not a page), the single day (the Workout planner’s pods, paces, and weather/terrain adjustments), and the athlete preview (Athlete day as a “preview” button, not a destination).

- **Text-to-structured-workout parsing** is the load-bearing technical piece: “2 mile wu, 5 tempo, 3 cd” typed in the grid must become a structured workout that can be analyzed post hoc.
- **Race plans** move here from Team board and attach to meet dates: per-athlete goals plus predicted pace ranges, and eventually MCMC field simulations against expected competition.
- **Training paces** (the Pace cards content) surface here as links; the paces themselves live on the athlete page.

Editing a day in place. Clicking a day — “Mon 7” — opens that day for editing directly, without leaving the week (Figure 4). The pop-up holds the text entry, the structure the parser made of it, and the **variants by attribute**: the same session prescribed differently to different groups. “5 tempo” for varsity boys, “4 tempo” for varsity girls and JV boys, “3 tempo” for JV girls is one workout with four variants, not four workouts, and the coach writes it once. Variants key on roster attributes (gender, varsity / JV, pod — see the Team section), and pods for the day are

Medium widget (M)	Shows	Type
Today's plan (compact)	the session by group, with print	text
Conditions strip	the four windows	text + icons
Needs a look (list)	the names behind the glance count, each linked	list
Most recent session	who did what against plan, in a paragraph	text
Upcoming meets	the next three: date, meet, entries due	list
Tasks	the organizer	list
Team weekly mileage	eight-week sparkline, planned vs. actual	chart
Pace groups today	who runs with whom, at what pace	list
Training load	acute:chronic ratio per group, as bars	chart
Weather trend	48-hour temperature and precipitation line	chart
Recent PRs	PRs from the last meet	list
Inbox	athlete messages and notes since yesterday	list
Large widget (L)	Shows	Type
Weekly mileage with CIs	team median weekly volume with an IQR band, planned volume overlaid; per-group toggle	chart
Session review	the most recent workout, prescribed vs. actual splits per athlete	table + chart
Week at a glance	Mon–Sun plan strip, today highlighted	calendar
Attendance heatmap	athlete $\times$ day, last four weeks	heatmap
Readiness grid	athlete $\times$ (sleep, HRV, load) as traffic lights	grid
Last meet results	team score, top finishers, PR count	table
Planned vs. actual pace	one dot per athlete for the last workout, with a CI band	chart
Training load timeline	team ACWR over the season, injuries marked	chart
Next race preview	course profile and the forecast for the race window	chart + text
Progression	selected athletes' 5k-equivalent over the season	chart

Table 3: Medium and large widgets. Chart widgets are rendered by the product, never assembled by the coach.

proposed from the pod settings: the coach can force the number of pods, the number of athletes per pod, and whether pods are mixed- or single-gender; the product fills them from training paces.

Mon, Sep 7 — edit day

2 mi wu, 5 tempo, 3 cd

parsed: wu 2 mi · tempo 5 mi @ T pace · cd 3 mi (10 mi)

variants by attribute

group	prescription	pods	athletes
varsity boys	2 wu · 5 tempo · 3 cd	A, B	14
varsity girls	2 wu · 4 tempo · 3 cd	C	7
JV boys	2 wu · 4 tempo · 3 cd	D, E	12
JV girls	2 wu · 3 tempo · 3 cd	F	6

Pods · proposed from training paces

Pods: auto

per pod: ~6

single-gender

Save

Preview athlete view

Push to athletes

one workout, four variants

Figure 4: Editing a day in place. One text entry, one parse, and a variant per attribute group; pods are proposed from training paces under the coach’s pod settings (count, size, gender mixing).

Build order: the four-zoom shell around the existing planner first; the text parser next; in-place day editing with attribute variants; the race-plan predictive layer last.

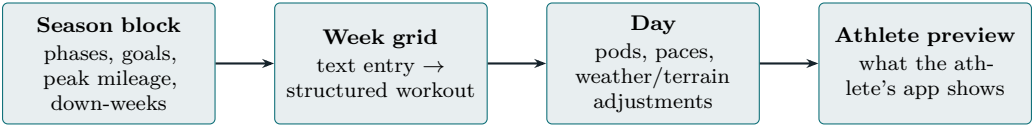


Figure 5: Upcoming: one planning object at four zoom levels. The current Planner / Workout planner / Athlete day split becomes zooming, not separate pages.

Figure 6 is the week zoom as a coach would see it: the zoom selector across the top, a text entry per day with the parsed structure shown beneath it, the race day highlighted, and the block-level context (week of the block, volume target, race plan status) in a rail below.

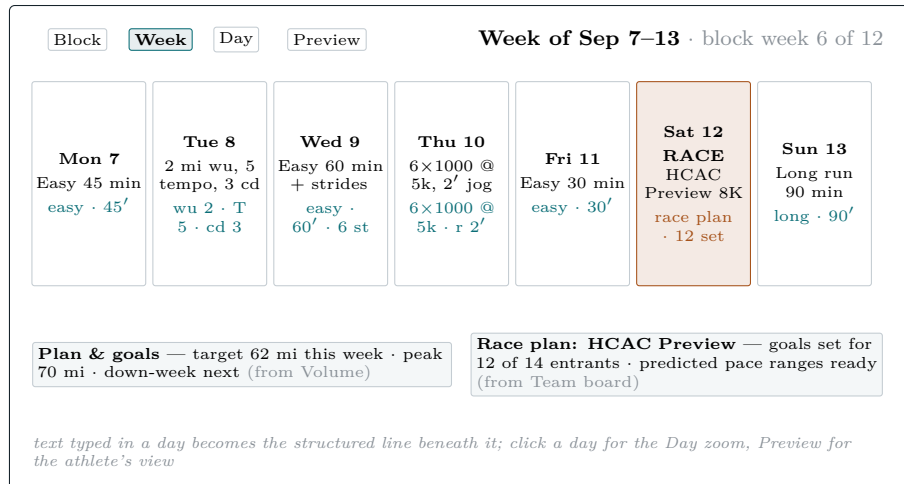


Figure 6: Upcoming, week zoom. Each day’s free-text entry is parsed into the structured workout shown under it (the load-bearing technical piece); the race day carries its race-plan status; the rail holds the block context.

Team

A team landing page — name, picture, conference/division information, the team document (rules / creeds / values), and the PR board — with four routes (Figure 7): the Roster, the Results repository (import as a pop-up, not a tab), the PR / season-progression boards, and Operations (money, equipment, the .ics export of the meet and training calendar). Every surface that shows an athlete’s name — roster row, result line, PR board entry — opens the same athlete page.

Roster attributes. The roster carries, per athlete, the attributes the rest of the product keys on: **gender**, **varsity** / **JV**, and **pod** (the training-pace group). Pods are either set by hand or proposed: the coach chooses the number of pods, or the number of athletes per pod, and whether pods are mixed- or single-gender, and the product estimates current pods from training paces and recent race marks. Workout variants in Upcoming address these attributes directly (“varsity girls and JV boys: 4 tempo”), which is why they live on the roster rather than in a settings page.

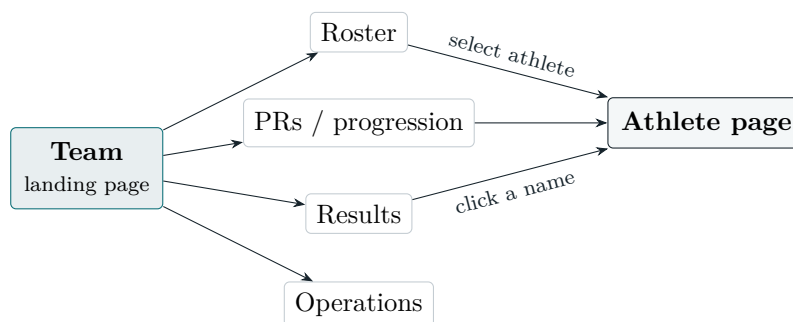


Figure 7: The Team tab. The athlete page is secondary: reached from the roster, the PR / progression boards, or any result line — all three converge on the same object. Operations stays team-level.

Figure 8 is the landing page with Rose-Hulman Institute of Technology as the example: an identity header (mark, name, location, current season, head coach, conference and division), then

a stack of blocks — coaches, athletes, results, PRs, schedule, operations — each of which opens its route. The block order is the coach’s: the handles on the left move a block up or down, so a coach who wants the roster first and the staff list last gets exactly that.



Rose-Hulman Institute of Technology
 Terre Haute, Indiana
Cross Country · Fall 2026
 Head coach: Derick Lawrence
 HCAC · NCAA Division III

↑↓ **Coaches**
collapse ^

Name	Title	Email	Phone
Derick Lawrence	Head Men's & Women's Cross Country / Assistant Track & Field Coach	lawrenc1@rose-hulman.edu	(812) 877-8246
Shane Mills	Head Men's & Women's Cross Country Coach	mills2@rose-hulman.edu	(812) 877-8051

↑↓ **Athletes** · 77 on the roster · 56 men, 21 women
 Roster →

↑↓ **Results** · last: NCAA Division III Championships, Nov 22, 2025
 Results →

↑↓ **PRs / progression** · record board, season progression
 Boards →

↑↓ **Schedule** · next meet, entries due, .ics
 Schedule →

↑↓ **Operations** · money, equipment, travel
 Operations →

Figure 8: The Team landing page, Rose-Hulman as the example. The Coaches block is shown expanded; the rest are collapsed rows that open their route. The ↑↓ handles reorder blocks. Roster counts are the 2025 XC season as held in the data platform.

The athlete page (three tiers, top to bottom).

Identity (top, glanceable).

Name, gender, class, hometown, MileSplit and Athletic.net links, and PRs — the public-roster feel the audit of the current Athlete page calls for.

Secondary.

Recent race results, recent training (a week-over-week mileage graph), attendance / training log, past and current injuries, upcoming registered events (date, name, event), training paces, and stated goals. Person-file material (dietary restrictions, class schedules, paperwork) stays populated further down; anything duplicated elsewhere (e.g. 1-on-1 notes) becomes a link, not a repeated widget.

Advanced analytics (bottom).

The learned attribute models:

- **Hill Resilience** — is the runner better on flat or hilly courses? Built from course-elevation learning, MileSplit course rankings, FloRatings A/B/C/D, TullyRunners speed ratings, joshm.dev’s Crystal Springs Pace Factor, and the golf-slope-rating analog.
- **Weather Impact** — how much adverse weather degrades this runner. The constraint: exact per-race time is needed, and getting it at scale from Athletic.net is not viable (see §2).

- Additional analytical models as desired.

Figure 9 is the page for a real athlete — Billy Allen, a Rose-Hulman sophomore — built from what the platform already holds: headshot and bio from the athletics site, collegiate results from TFRS, the high-school career from MileSplit (his profile found by name and class year), and the course-adjusted numbers from the ALS fit of §2. The MileSplit and Athletic.net links are badges in the sites’ own marks so a coach recognizes them at a glance.

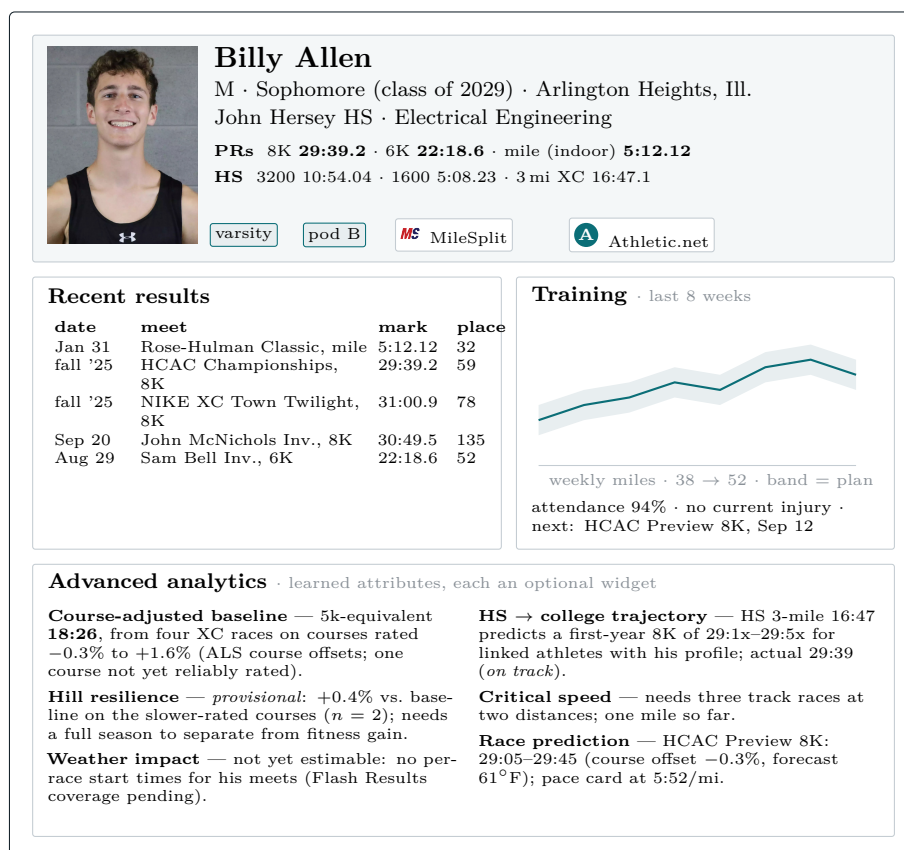


Figure 9: The athlete page for Billy Allen (Rose-Hulman, class of 2029): identity tier with PRs and the HS career via the MileSplit link; recent results and training in the secondary tier; the learned attributes at the bottom, each stating what it needs when it cannot yet be estimated. Bio, PRs and results are real (two meet dates are absent from the TFRS pages themselves); the training card, attribute chips, and analytics values are illustrative of the output format.

Analytics that could be added. Beyond hill resilience and weather impact: a **race-prediction band** per upcoming meet from the baseline, course offset and forecast; **critical speed** / D' from track marks; **championship response** (does the athlete run above baseline at championships); **negative-split tendency** from split data where timing provides it; **training response** — the baseline’s movement against mileage and workout load, which is the first place the GetFast training data joins the results data; **consistency** (race-to-race variance, which is itself a selection signal); and the **HS → college trajectory** described in §2, which places a first-year athlete against the linked population with the same high-school profile.

Build order: the identity and secondary tiers ship first — they display data already held. The analytics tier lands as the platform models mature (§2), each model appearing as an optional widget

rather than a blank promise.

Recruiting (v2)

Reserved as a tab but not built in v1; recruit-visit mode (masking athlete names) ships in v1 under Settings. When v2 comes, the tab inherits the recruiting file from the current person file, and the HS ↔ college linkage of §2 is its natural engine: a coach looks up a prospect’s verified HS results and course-adjusted marks rather than a self-reported PR.

Settings

One standardized menu (Figure 10) replaces the Housekeeping tier: data importation from Data health, the Setup panels split into their own tabs, the Workout planner’s “Rules baked in,” the Parameters and Predictions pages, feature toggles (the grand-strategy layering — run the product as anything from roster-plus-planner up to the full analytics surface), recruit-visit mode, and bulk import/export, with “How this works” as the help entry. If the paperwork tracked in Data health is medical clearances, it moves to the Team tab’s roster as a who-is-missing-paperwork widget — it is compliance, not data flow. This tab is a mechanical regrouping with no new functionality, which makes it a good early milestone.

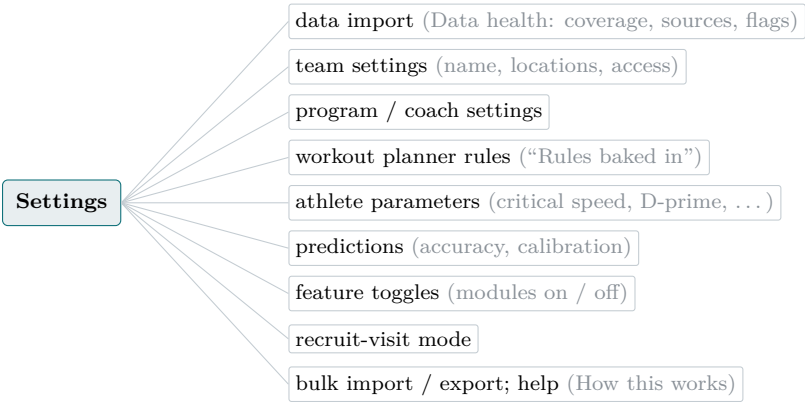


Figure 10: The Settings menu, centralizing what is currently scattered across Housekeeping, the Workout planner, and the athlete sub-pages.

1.5 Wording

Much of the copy reads as too cute, too artsy, or too AI-generated, and should be plainer. Examples:

Current	Plainer
“One line from an athlete at 3:47 is worth more than finding out at 4:10 in the parking lot.”	Cut, or state the actual point.
“Where your attention goes” — “ranked, and every name is a door.”	“Needs a look,” with names that link.
“Yesterday, rolled up”	“Yesterday, summarized.”

2 Data and Enrichment Platform

The athlete attributes in §1 (hill resilience, weather impact, and the models to come) are only as good as the data feeding them. This section lays out how that data is sourced, enriched, and linked — and why the platform must **divest fully from Athletic.net** as a scale dependency.

2.1 (1) Roster, meet, and team information

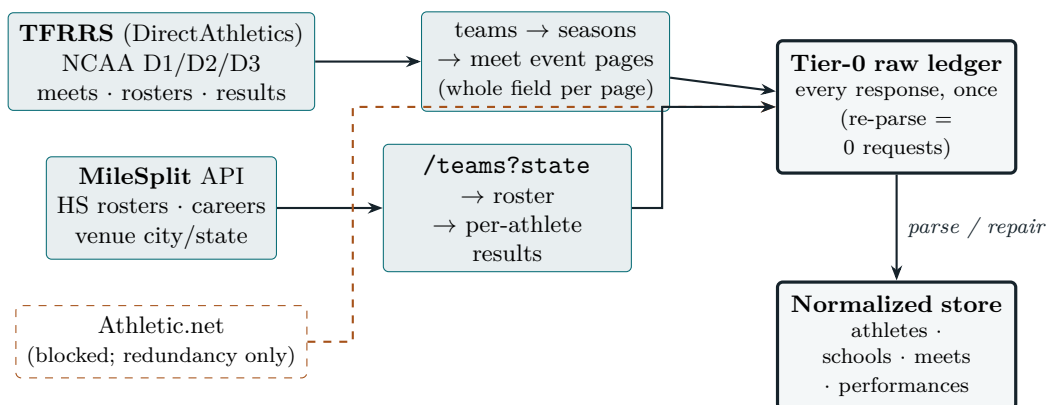


Figure 11: Roster/meet/team sourcing. Every response tees to a raw ledger before parsing, so a parser fix replays at zero request cost. Athletic.net (dashed) is demoted to redundancy.

How it works, per source.

- **TFRRS (college).** Seeded from the seven NCAA-division region pages, walk teams → seasons → meets. A single meet *event* page carries the whole field (place, athlete, class year, team, mark), so one fetch covers every school in a race. Corpus: 8.1M rows, all three divisions, XC + indoor + outdoor, 2021–2026.
- **MileSplit (HS).** `/teams?state=xx` then the proven per-team, per-athlete ingest path, smallest state first. Polite by default (rate-limited, ledger tee). Note the enumeration cap: `/teams?state` returns at most 999 rows, so the largest states need a per-county or meet-harvest route for full coverage.
- **Divest path.** Athletic.net has the broadest US high-school coverage and the project was built on it, but it is not a dependency to scale on: Cloudflare returns `cf-mitigated=challenge` on sustained crawling (1–2 req/s ran clean for days; 3 req/s drew 24-hour blocks on consecutive days; exit-node rotation did not help — the block is on the traffic pattern, not the address), there is no API, developer program or partnership channel, and its rows carry no race times and collapse relays to one athlete per squad. So: new states come from MileSplit; the existing Athletic.net captures (CA/TX/FL/NY) are retained and merged as a redundancy check; no new pipeline depends on that host.

2.2 (2) Course weather and hill information

Wind is an essential characteristic of a sprint mark. A 10.90 at +3.0 and a 10.90 at −1.0 are different performances, and the rule book agrees (marks over +2.0m/s are not record-legal). For the wind-legal events — 100, 200, the hurdles, long and triple jump — the measured gauge reading is stored on the result row itself (Athletic.net carries it on ~9% of track rows; TFRRS

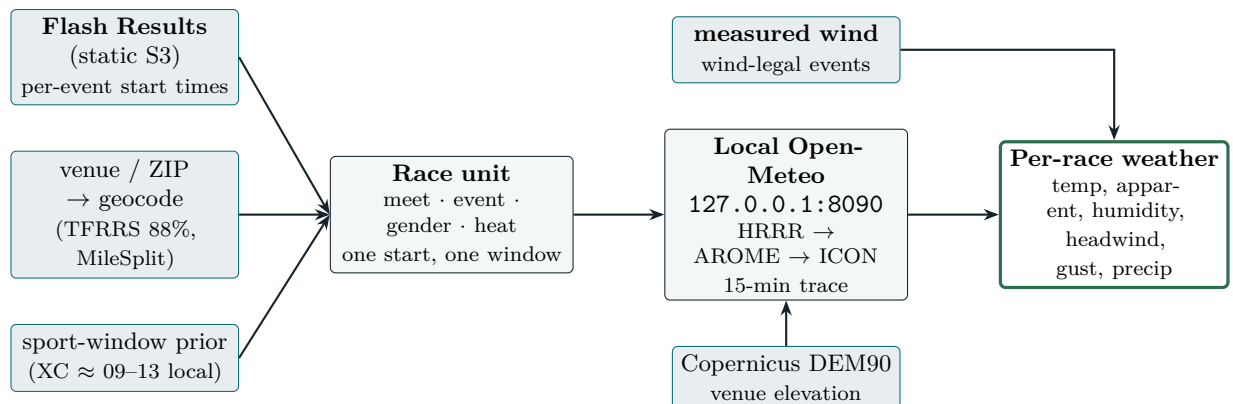


Figure 12: Weather + elevation per *race*, not per meet. The 11 am women’s JV and the 3 pm men’s varsity get different windows. Wind for sprints comes measured from the results themselves.

event pages carry a WIND column), and every sprint analytic is conditioned on it. Forecast wind from the weather model is the fallback for distance events, never a substitute for the gauge on a sprint.

Hill / course difficulty (three complementary signals):

1. **Empirical ALS course offsets (built).** A two-way log-time model — $\log(\text{equiv5k}) = \text{ability} + \text{offset}$, i.e. an athlete-season term plus a course term. **ALS** is *alternating least squares*: hold the course offsets fixed and solve each athlete’s ability as the least-squares (mean) fit to their races; then hold abilities fixed and solve each course’s offset the same way; alternate until nothing moves (400 rounds), with the offsets anchored to mean zero so the split is identified. It needs only athletes who race more than one course. Validated split-half $r = 0.98$; recovered Crystal Springs (+72s) and Woodbridge (−50s) with no external input. It measures what terrain *costs*, but bundles hills with footing and weather.
2. **External course ratings (cross-check + decomposition).** TullyRunners speed ratings, MileSplit course rankings, FloRatings A/B/C/D, joshm.dev Crystal Springs Pace Factor, golf-slope analog. Joining these and correlating with the ALS offset validates our number and separates terrain-driven from conditions-driven difficulty.
3. **True elevation gain (targeted).** GPS/GPX course traces through the DEM give cumulative climb per km — the only physical hill metric, but per-course external data. Reserve for marquee courses.

Held-out test on 100 athletes per division. To check that the offsets carry information rather than fit noise, 100 athlete-seasons with at least three XC races were drawn at random from each of D1, D2 and D3 (seed 42), one race per athlete was held out, and the model was re-fit on the remaining 619,056 rows (151,239 athlete-seasons, 3,410 meets). For each held-out race the question is whether the *course offset* explains what the athlete’s own baseline cannot (Figure 13, Table 4): the correlation between the athlete’s deviation from baseline and the course’s offset is 0.50–0.76, the prediction error falls 20–36% once the offset is applied, and a permutation null (offsets shuffled across the held-out races, 2,000 draws) never reaches the observed correlation ($p < 0.0005$, null s.d. 0.10). D2 is the weakest, consistent with its thinner meet overlap.

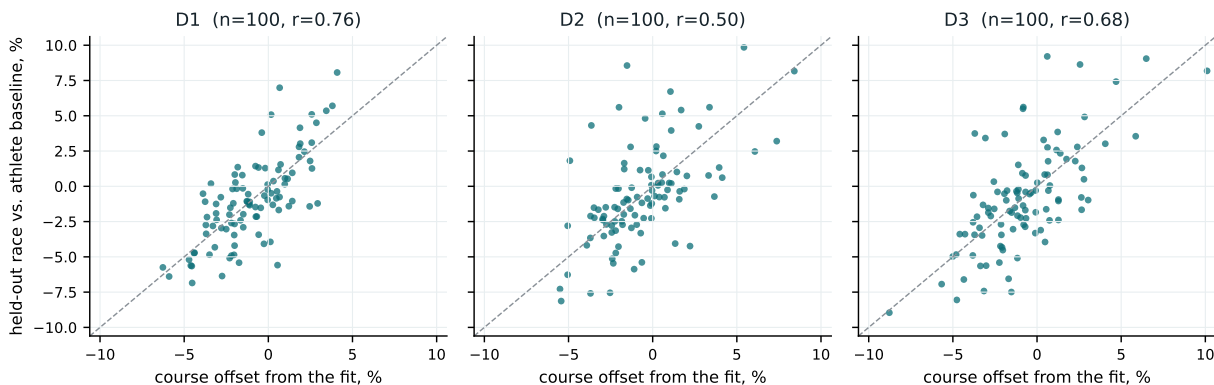


Figure 13: Held-out race vs. course offset, 100 random athletes per division. Each point is one held-out race: x is the course’s fitted offset, y is how far the athlete ran from their own baseline that day. Points on the dashed line are fully explained by the course.

	n	$r(\text{dev.}, \text{offset})$	r pred.	r base	MAE base	MAE w/ offset
D1	100	0.76	0.98	0.95	28.4 s	18.2 s (−36%)
D2	100	0.50	0.97	0.97	35.4 s	28.3 s (−20%)
D3	100	0.68	0.98	0.97	35.3 s	24.1 s (−32%)

Table 4: Held-out validation of the ALS course offsets. MAE is on the 5k-equivalent time. Permutation $p < 0.0005$ in every division.

Two caveats found while running it. The fit was made on the corpus as it stands, which still holds 2,705 duplicate XC rows across 46 meets from the July legacy sync (607 with a conflicting distance label) — small, and a targeted delete, but one affected meet’s offset is wrong until it is done. And 259k XC rows belong to schools whose TFRRS slug did not resolve to a division; the samples are drawn from the 405k that did.

Assessment. Once weather is attached per race, re-fit the ALS *net of the weather term*: the course offset then stops absorbing that day’s heat and becomes a terrain-only score — the cleanest hill metric derivable without a GPS trace.

2.3 (3) Linking college and high-school runners

There is no crosswalk: zero athletes carry both an Athletic.net id and a TFRRS id, and MileSplit’s `tfrsId` field is null on every bio examined. Linkage is therefore *probabilistic*, and the honest deliverable is the ceiling.

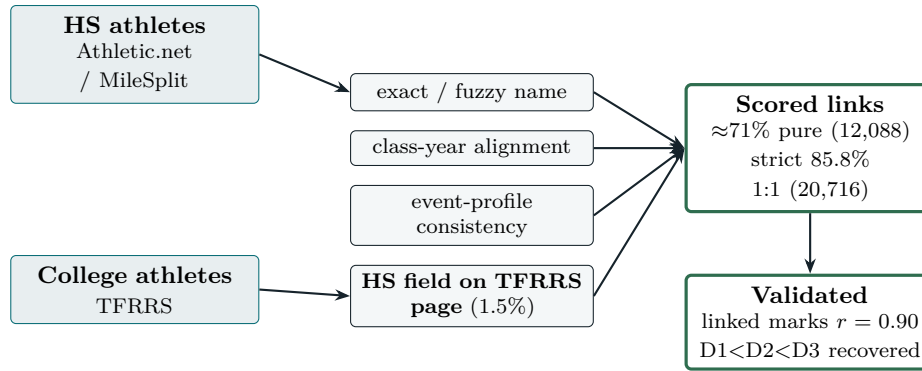


Figure 14: Probabilistic HS→college linkage. No exact key exists; a scored combination of name, year and event profile reaches $\sim 71\%$ precision, and TFRRS athlete pages carry a literal High School field (1.5% of pages) that opens a path to a *real* crosswalk by crawling those pages directly.

Why link at all. The link is what turns two separate result histories into a *trajectory*. With it the platform can say how athletes with a given high-school profile developed in college — expected first-year 8K from a 3-mile PR, typical improvement by event and by division, who bloomed late — and place any current athlete against that population: Billy Allen’s 16:47 3-mile became a 29:39 8K as a freshman, and the linked population says whether that is ahead of, on, or behind the curve. That **potential-growth** view is the core of recruiting (v2): a prospect’s verified HS marks projected forward, rather than a self-reported PR read at face value. It also benchmarks a program (what HS profile did our athletes who succeeded arrive with) and lets course-adjusted HS and college marks be compared on one scale.

The route to a real crosswalk. TFRRS athlete pages carry a literal “*High School:*” field on 1.5% of archived pages (roster pages do not). Crawling athlete pages directly — rather than inferring identity from names — upgrades the linkage from probabilistic to keyed for the athletes who have it, and is the recommended next step for the recruiting and progression models.

2.4 (4) Finding meets to enter

A coach’s recurring question — what is coming up, and when do entries close — has a scalable answer for high school and an open one for college.

- **High school: MileSplit’s meet feed.** The meets endpoint, `/api/v1/meets` with a `state` parameter, returns a rolling window of roughly 60 days per state with `dateStart`, `dateEnd`, `season`, venue city/state and a `registrationActive` flag (probed 2026-08-22: 100 Indiana meets, 65 with registration open). It has no deadline field and ignores `page`, so the pattern is: poll each state once a day (51 requests), diff against what was seen, and fetch the meet’s own page once for any *new* meet with registration open to read the entry deadline. Filter by distance from the team’s geocode, by level, and by season, and the result is an “upcoming meets near you, entries due by” list that costs a few hundred requests a day nationally.
- **College: the team’s own schedule page.** TFRRS lists only past meets, and DirectAthletics — where college entries actually happen — answered our client with a CloudFront 403. But college athletics sites run on a handful of templates (Sidearm, PrestoSports), and Rose-Hulman’s served us without challenge: the schedule page carries every scheduled meet with date, host and an `.ics` export. Reading a team’s schedule from its athletics site is uniform across the template

and is the college route; entry deadlines still come from the coach’s DirectAthletics account or the host’s meet page.

2.5 Build order

Steps that cost nothing against any host come first:

1. Venue + geo columns; reparse venue from the archived TFRRS/MileSplit payloads. **Zero requests.**
2. Wind column on performances; backfill from archived Athletic.net payloads (wind-legal events). **Zero requests.**
3. Flash Results time crawler — new host, static S3, its own polite profile and ledger. Start with the NCAA championship/regional set and measure the match rate before scaling.
4. Weather adapter: emit a per-race manifest, run the existing `~/weather` Open-Meteo pipeline unchanged, reduce to race-window aggregates.
5. Course difficulty: ship the ALS offset, join external ratings as a check, then re-fit net of weather for a terrain-only score.

2.6 v2 — Recruiting: considerations and goals

Recruiting is deliberately a v2 product: it needs the linkage and the trajectory models above to be real before it is more than a contact list. Goals, in order:

1. **Prospect lookup with verified marks.** A name resolves to the athlete’s MileSplit / Athletic.net record, and the page shows their marks as the platform holds them — course-adjusted, wind-conditioned where it applies — not as reported.
2. **Potential growth.** The HS → college trajectory placed against the linked population: expected first-year marks, the range, and the comparable athletes it is drawn from, by division.
3. **Fit to program.** Where the prospect would sit on this roster (varsity / JV line, pod) and against the conference, using the same projections as the Team tab.
4. **Watch lists and timelines.** Prospects by class year with contact-period dates, the recruiting file that today lives in the athlete person file, and notes.
5. **Recruit-visit mode** stays in v1 under Settings: masking athlete names on a campus visit is a display toggle, not a recruiting feature.

Considerations: NCAA contact rules vary by division and must be encoded, not assumed; high-school data comes from sources with redistribution terms (MileSplit’s non-commercial etiquette, Athletic.net’s ToS), so a commercial recruiting product needs a licensing conversation before it ships; and the linkage is probabilistic (~71% pure today), so every projection must carry its confidence and its comparables, never a bare number.